

TABLE OF CONTENTS

SUBSECTION-4.2 MECHANICAL WORKS

	Page. No.
SUBSECTION 4.2 MECHANICAL WORKS	4
4.2.1. CLEANING DURING AND AFTER ERECTION	4
4.2.1.1 Equipment and Piping	4
4.2.1.2 Cut-Ins and Openings	4
4.2.1.3 Closures	4
4.2.1.4 Valves and Other Specialties	4
4.2.1.5 Cleaning Materials and Procedures	4
4.2.1.6 Cleaning of Lubricating Oil Systems	4
4.2.1.7 Cleaning of Bearings	5
4.2.2. FIELD ROUTING	5
4.2.2.1 Piping	5
4.2.3. PIPE SUPPORTS	5
4.2.3.1 General	5
4.2.3.2 Supplementary Steel	5
4.2.3.3 Adjustment and Testing	6
4.2.3.4 Temporary Supports	6
4.2.4. PIPING ERECTION	7
4.2.4.1 Minimum of Joints	7
4.2.4.2 Diagrammatic Connections	7
4.2.4.3 Alignment and Drainage	7
4.2.4.4 Closures	7
4.2.4.5 Clearance for Insulation	7
4.2.4.6 Attachments for Insulation, Pipe Supporting Elements and Handling ..	7
4.2.4.7 Cutting, Drilling etc. of Building Structure	8
4.2.4.8 Floor, Wall and Roof Openings	8
4.2.4.9 Flanges and Gaskets	8
4.2.4.10 Installation of Studs for High-Pressure Flanged Joints	8
4.2.5. FIELD WELDING	8
4.2.5.1 General	8
4.2.5.2 Non Destructive Examination (NDE) and Testing	9
4.2.5.3 Fit Up and Alignment	9

4.2.5.4	Minimum Wall Thickness	9
4.2.5.5	Stamping	9
4.2.5.6	Testing of Welds.....	9
4.2.5.7	Welding Procedures and Qualification	9
4.2.5.8	Alloy Piping Welds.....	10
4.2.6.	CONNECTION TO EQUIPMENT	10
4.2.6.1	General	10
4.2.7.	PIPING PROTECTION	10
4.2.8.	FIELD TEST FOR PIPING SYSTEM	11
4.2.8.1	Scope of Work.....	11
4.2.8.2	Test Requirements	11
4.2.8.3	Hydrostatic Test.....	11
4.2.8.4	Test Pressures and Temperatures	11
4.2.8.5	Pneumatic Tests.....	12
4.2.8.6	Preliminary Leak Test.....	12
4.2.8.7	Leak Testing on Buried Piping.....	12
4.2.8.8	Leak Testing on Encased Piping	12
4.2.8.9	Testing After Repairing	12
4.2.8.10	Temporary Field Testing Facilities and Services	13
4.2.8.11	Gagging During the Leak Test	13
4.2.8.12	Run Test	13
4.2.8.13	Adjustment and Tightness.....	13
4.2.9.	INSULATION	14
4.2.9.1	General	14
4.2.9.2	Materials.....	14
4.2.9.3	Multiple Layer Insulation.....	14
4.2.9.4	Insulation Fastening Wire	15
4.2.9.5	Joints	15
4.2.9.6	Insulation for Hot Piping.....	15
4.2.9.7	Insulation for Hot Flanges.....	16
4.2.9.8	Insulation for Hot Valves and Fittings.....	16
4.2.9.9	Removable Covers for Valves and Fittings	16
4.2.9.10	Insulation for Hot Equipment.....	16
4.2.9.11	Personnel Protection	17
4.2.9.12	Insulation for Temporary Piping	18

4.2.9.13	Cladding	18
4.2.9.14	Canvas Jacketing	18
4.2.9.15	Weatherproof Coating and Jacket.....	19
4.2.10.	FLUSHING AND STEAM BLOW	20
4.2.10.1	Piping and Vessels	20
4.2.10.2	Feedwater Systems	20
4.2.10.3	Oil Systems	21
4.2.10.4	HRSG (except for Superheaters, Steam Piping and Pumps).....	21
4.2.10.5	Superheaters and Steam Piping - Steam Blow	22
4.2.10.6	Disposal of Waste Effluent	23
4.2.10.7	Inspection after Cleaning	23
4.2.10.8	Submittal for Cleaning Procedure	24
4.2.11.	FIELD TESTS FOR MECHANICAL EQUIPMENT	24
4.2.11.1	HRSG	24
4.2.11.1.1	Hydrostatic Test	24
4.2.11.1.2	HRSG Setting Smoke Test	24
4.2.11.1.3	Storage of HRSG.....	24
4.2.11.2	Field Fabricated Tanks.....	24
4.2.11.3	Condensing Plant	25
4.2.11.4	Air-Conditioning and Ventilations	25
4.2.11.5	Fire Protection Services	27

SUBSECTION 4.2 MECHANICAL WORKS

4.2.1. CLEANING DURING AND AFTER ERECTION

4.2.1.1 Equipment and Piping

The Bidder shall ensure that all mechanical equipment erected by him is kept clean during and after erection. The piping cleaning shall be in accordance with the requirements of **Part 4, Technical Specification**.

4.2.1.2 Cut-Ins and Openings

The Bidder shall clean out the inside of all cut-ins and openings made by him in the piping for new connections. The Bidder shall blow out the piping prior to making new connections, if required.

4.2.1.3 Closures

All openings shall be provided with temporary closures, as soon as possible after cleaning. Also if new connections to cut-ins are not made immediately the openings shall be sealed with temporary protectors.

4.2.1.4 Valves and Other Specialties

Valves and other specialties and fabricated piping furnished and erected by the Bidder shall be shop-cleaned prior to shipment. However, the Bidder shall inspect the inside of such materials, upon receipt and prior to installation, to ensure that they are clean. The Bidder shall reclean such materials, if necessary.

4.2.1.5 Cleaning Materials and Procedures

Prior to start of the Works the Bidder shall submit his proposed methods and procedures for cleaning the piping. to Owner for review.

4.2.1.6 Cleaning of Lubricating Oil Systems

The Bidder shall thoroughly clean and flush out all lubricating oil systems and shall afterward fill systems with lubricating oil. Flushing oils, cleaning medium and lubricating oils shall be furnished by the Bidder. The Bidder shall furnish any heating devices required during flushing of the lubricating systems. The final flushing procedure shall be on a continuous around the clock basis. The fulfillment of the above requirements shall be carefully checked, and subject to acceptance by the Owner. The Bidder shall thoroughly clean and flush the oil purification system.

4.2.1.7 Cleaning of Bearings

The Bidder shall thoroughly clean all bearings after which he shall fill bearings and bearing lubricators with oil or grease as and where required. Flushing oil (if required) and lubricants shall be furnished by the Bidder. The Bidder shall remove and replace bearing covers and such other work as may be necessary for bearing inspection after the initial trial run of the equipment.

4.2.2. FIELD ROUTING

4.2.2.1 Piping

Routing of small piping etc. shall be fixed at Site. Extreme care and judgment shall be exercised in routing small pipe, tubing, conduit etc. which is shown only diagrammatically on the Bidder's design drawings, so as to provide sufficient clearance under and/or around hatchways, galleries, monorails, removable slabs, temporary end walls, electrical switchgear, gauge boards etc. for maintenance, operation, inspection and the like. All piping with a diameter over DN25 shall be shown in the piping diagrams. Valves

All field located valves shall be arranged for accessibility. Instrument valves, drain valves and vent valves shall be accessible from galleries and floors without plant operators having to reach out, up or down any more than necessary. Where there are double valves both valves shall be located so that they are accessible from the galleries or floor.

4.2.3. PIPE SUPPORTS

4.2.3.1 General

The erection of the pipe supports shall be in accordance with the applicable requirements of the **Part 4, Section 3, M-002** and other requirements specified herein. Supports for the field routed piping shall also be in accordance with **Part 4, Section 3, M-002**.

The Bidder shall design and locate all piping supports required for field-fabricated and/or field-routed piping, unless otherwise indicated, including any special supports required for small piping.

Outdoor instrument piping and tubing supported from structural steel shall be arranged in a suitable manner so as to prevent excessive heat loss to the steel members.

All braces that may be required to eliminate piping vibration shall be erected by the Bidder to the full satisfaction of the Owner.

4.2.3.2 Supplementary Steel

Bidder shall design and furnish all supplementary steel as required.

4.2.3.3 Adjustment and Testing

The Bidder shall ensure that all the temporary locking devices or stops are engaged and thus maintained on all constant and variable-support spring units, until the piping systems are placed in service, except during such time of checking and making adjustments on the supports. The temporary locking devices shall be engaged during filling, testing and draining of the piping systems for hydro test.

The Bidder shall ensure the following tasks are completed after erection of piping and associated supports but before preliminary operation startup:

- Adjust all supports for proper tension or loading and set all travel indicators to cold mark.
- Check and, if required, adjust the travel stops on spring-type units.

After startup of piping systems and normal operation has been attained, check all supports for proper tension (or loading) and movement, and check that all travel indicators are on hot mark setting.

After shutdown from preliminary operation and after piping systems have returned to cold condition, check supports and record any changes in cold setting from those specified above.

Examine each restraint and guide, in both the cold and hot positions, to verify that pipe can move freely, and adjust if required. Where restraints or guides cannot be observed in the hot operating condition, actual measurements of the cold position of the pipe with respect to the restraint or guide shall be recorded. These measurements shall be compared with those shown on the design drawings for the purpose of verifying the adequacy of clearance for piping movement.

Prior to startup and operation of the piping systems, remove all temporary locking devices or temporary stops on all spring type units and attach them to their units.

Inspect the final setting of each support, record and certify as to meet the requirements of the applicable codes and Specification. An affidavit shall be submitted to Owner attesting to these requirements.

4.2.3.4 Temporary Supports

The Bidder shall furnish and install temporary supports or bracing as required during hydrostatic testing or leak testing, to prevent undue stress on the systems permanent piping supports and to withstand the weight of water without excessive deformation.

The Bidder shall remove any/all temporary supports, anchors and bracings after final and successful completion of field hydrostatic and leak testing.

4.2.4. PIPING ERECTION

Piping erection including valves, pipe, supporting elements and appurtenances, shall be erected in an orderly and workmanlike manner and shall be scheduled to suit the requirements of the Project. In addition to other applicable requirements specified herein, the following specific piping system erection requirements shall apply.

4.2.4.1 Minimum of Joints

Throughout the entire systems, there shall be as few field joints as possible.

4.2.4.2 Diagrammatic Connections

Any connections which are indicated only diagrammatically on the design drawings shall be installed in a neat and workmanlike manner, subject to acceptance by the Owner.

4.2.4.3 Alignment and Drainage

Piping shall be erected so as to preserve accurate alignment. The Bidder shall check field measurements and allowance for makeup lengths or “closures” as may be necessary for accurate alignment and assembly, prior to fabrication and installation.

4.2.4.4 Closures

All piping shall be worked into place and the closing weld made with a minimum of springing or forcing. It shall be the Bidder’s responsibility to maintain surveillance of the piping system during erection to ensure that no stress from misalignment is imposed, and that dimensional requirements are met. This should include a review of the piping support in the field to ensure that they are properly located and adjusted to the design position. Neutral points shall be maintained.

4.2.4.5 Clearance for Insulation

Piping shall be installed with suitable clearance being allowed for insulation where pipes cross or run in parallel, or where pipes run close to equipment or building structures.

4.2.4.6 Attachments for Insulation, Pipe Supporting Elements and Handling

Where attachments to piping are required such attachments shall be in accordance with the applicable requirements of **Part 4 Section 3, M-001** of Piping Specification.

4.2.4.7 Cutting, Drilling etc. of Building Structure

The Bidder shall do all necessary cutting and patching of concrete and brickwork. Under no circumstances shall the Bidder do any such cutting without the written consent of the Owner.

The Bidder shall furnish anchoring facilities. Where details are not shown on the drawings, the Bidder shall submit details to the Owner for review and acceptance.

4.2.4.8 Floor, Wall and Roof Openings

Floor, wall and roof pipe openings and sleeves and roof opening curbs required in the building structure shall be provided by the Bidder. Piping-routed by the Bidder shall be run in groups, insofar as practical, to minimize the number of pipe openings, sleeves and guarding devices required.

The Bidder shall install the guarding devices for the pipe sleeves and pipe openings. After all piping is erected, the Bidder shall provide additional guarding devices, if required.

The Bidder shall install the flushing points as required.

4.2.4.9 Flanges and Gaskets

Faces of flanges and gaskets shall be wiped clean prior to making up all flanged joints.

Contact faces of all flanged joints shall come squarely together and particular care shall be exercised in pulling up flanged joints to prevent any overstressing of flanges, bolts or studs.

No permanent fillers shall be used between flanges, except where specifically indicated on the design drawings and approved by the Owner.

4.2.4.10 Installation of Studs for High-Pressure Flanged Joints

All studs shall be installed in accordance with of **Part 4 Section 3, M-001** Piping Specification.

4.2.5. FIELD WELDING

4.2.5.1 General

Field Welding shall in general be in accordance with the requirements of **Part 4 Section 3, M-001**, Piping Specification in addition to the requirements specified herein. Field welding of galvanized piping is prohibited.

4.2.5.2 Non Destructive Examination (NDE) and Testing

All Non Destructive Examination (NDE) and testing shall be per the applicable requirements of **Clause 6 of Part 4 Section 3, M-001**, Piping Specification.

4.2.5.3 Fit Up and Alignment

All fit up and alignment shall be per the requirements specified in **Part 4 Section 3, M-001**, Piping Specification.

4.2.5.4 Minimum Wall Thickness

The minimum wall thickness specified or required for pipe, fittings etc. shall be maintained at weld joints, as well as at other areas.

4.2.5.5 Stamping

All welders qualified shall be provided with an identification stamp of the low stress type. The welder shall stamp each of his welds with his identification. The stamp shall be located on the base material adjacent to the weld.

4.2.5.6 Testing of Welds

If for any reason it becomes necessary, in the opinion of the Owner, to test the quality of welds that have been made by the Bidder in field-fabricated assemblies, standard test specimens shall be removed from the designated welds by the Bidder and tested in the presence of the Owner.

Field welding and stress relieving on piping and equipment shall be acceptable to the Owner. All stress relieving shall be done either by electrical induction or electrical resistance heating

4.2.5.7 Welding Procedures and Qualification

The Bidder shall be responsible for the welding performed by his organization and shall establish his own welding procedures for each class and type of weld that will be used in the Work.

Welders shall be qualified in accordance with Section IX of the ASME Boiler and Pressure Vessel Code and Indonesian Government Regulation.

Welders shall be qualified on the jobsite. Certification of all welder qualifications shall be supplied to the Owner.

Before any field welding is performed, the Bidder shall submit copies of his welding and stress relieving procedures to the Owner for their comments.

4.2.5.8 Alloy Piping Welds

Welding of alloy pipe joints shall be completed without interruption wherever practical.

All alloy piping field welds shall be radiographically examined in accordance with procedures of ASME Section V and acceptance standards of Paragraph PW51 of ASME Section I Power Boilers. The radiographic procedure shall be submitted to the Owner for review and comments. All radiographic film and stress relieving charts shall be submitted to the Owner.

4.2.6. CONNECTION TO EQUIPMENT

4.2.6.1 General

Connections to the equipment shall be made as indicated on the drawing and as required.

Where weld joints, rubber gasketed joints or mechanical couplings are used, the Bidder shall ensure that stress levels do not exceed code allowables.

All welds shall be continuous full penetration welds and shall be on both the front side and back side of the plates, unless otherwise approved.

4.2.7. PIPING PROTECTION

For piping of 50 mm in diameter and smaller field-fabricated by the Bidder, all openings of the fabricated piping and remaining cut pipe shall be provided with protectors, as soon as possible after cleaning.

If connections to piping erected by the Bidder are required, the Bidder shall ensure that the terminal connections are provided with protectors and that such protectors are as specified in the Specification. Where such terminals are located in an area where they may be subject to damage from other operations at the Site or constitute a hazard, the Bidder shall provide barricades and/or warning signs around the terminals.

The Bidder shall, at the end of each working day, ensure that all remaining openings in piping erected by him and all new openings made by him in existing piping are provided with temporary protectors.

Mechanical joints using thread system parts such as screws, nuts and bolts etc. shall not be coated in the shop but shall receive a suitable rust preventative only, and after final field assembly the joint shall be thoroughly cleaned and coated with the specified system.

Where rubber-gasketed joints or mechanical couplings are used, the shop-applied or field-applied coating shall not be in excess of that recommended by the sealing device or coupling manufacturer.

4.2.8. FIELD TEST FOR PIPING SYSTEM

4.2.8.1 Scope of Work

Prior to initial operation, the Bidder shall perform leak tests on each part of the system, unless such tests are specifically waived or otherwise specified by the Owner.

4.2.8.2 Test Requirements

All tests shall be conducted in accordance with the requirements of ANSI B31.1. - Section 1, Power Piping Code.

4.2.8.3 Hydrostatic Test

All systems including but not limited to those listed in **Part 4 Section 3, M-001** Piping Specification shall be given a hydrostatic test per ANSI/ASME B31.1.

4.2.8.4 Test Pressures and Temperatures

As set forth in ANSI/ASME B31.1 and as follows:

- Water used for the hydrostatic tests shall in no case be less than 20°C or greater than 50°C for all systems.
- The hydrostatic test pressures shall in no case exceed the adjusted pressure-temperature rating for 38°C as given in the ANSI Standard for Steel Pipe Flanges and Flanged Fittings (ANSI B16.5) and the ANSI Standard for Steel Valves (ANSI B16.34) for the material and pressure standard involved.
- Underground Services and Drains. All underground services and drains at atmospheric pressure shall at completion of routing, be tested hydrostatically under a head of not less than 3 metres of water for a period of 30 minutes.
- Water, Oil and gas Piping All water, oil and gas piping before being concealed or insulated and before acceptance of the work shall be subjected to a hydrostatic pressure test of one and one half times its design pressure but not less than 4.5 bar g a period of two hours.

All gas fuel piping supplied under this Contract shall be purged and inerted under this Contract to the Owner's approval prior to filling with gas fuel.

Particular attention is to be paid to the Site cleaning of lubricating oil pipes which shall be to the owner's approval.

- Air Piping
All air piping before being concealed and before acceptance of the work shall be subjected to an air test of one and one half times its design pressure but not less than 4.5 bar g and all joints shall be checked for

leaks using a soap solution test. Test shall be made for a period of two hours.

All compressed air lines shall be blown through with compressed air before connection its made to instruments and other equipment.

- Chlorine Piping. All chlorine piping before being concealed and before acceptance of the work shall be subjected to a hydrostatic pressure test of one and one half times its design pressure but not less than 4.5 bar g for a period of two hours.

On completion of the chlorine piping installation the system shall be dried and subjected to a gas leak test.

4.2.8.5 Pneumatic Tests

The following systems shall be given a pneumatic test in accordance with the ANSI/ASME B31.1 Code except for any ASME Boiler and Pressure Vessel Code portions. The test medium shall be nitrogen or carbon dioxide gas. The pneumatic test shall be preceded by a preliminary test per ANSI/ASME B31.1.

- Hazardous, toxic or flammable materials piping systems where, in the opinion of the Owner, the use of water or oil as a test medium could result in contamination.
- Where tanks or vessels are within the test boundaries, they shall be isolated from the piping system, or the pneumatic test shall be waived in favor of a preliminary leak test.

4.2.8.6 Preliminary Leak Test

A preliminary leak test shall be given per ANSI/ASME B31.1, with clean, dry and oil-free air, followed by an Initial Service Leak Test per ANSI/ASME B31.1.

4.2.8.7 Leak Testing on Buried Piping

Backfilling of piping buried in the earth may be done, prior to leak testing, at Bidder's risk. In such cases, where leaks are detected by excessive pressure drop in test pressure, by visual examination, or by any other means, the associated area shall be reexcavated and the leaks repaired at no additional cost to the Owner.

4.2.8.8 Leak Testing on Encased Piping

Encasement of piping in concrete shall not be done prior to leak testing, unless accepted by Owner.

4.2.8.9 Testing After Repairing

Where detected leaks are repaired, the area radiographed, and the radiograph evaluation accepted, hydrostatic or pneumatic leak test of the system need not be repeated.

4.2.8.10 Temporary Field Testing Facilities and Services

The Bidder shall furnish, and install all temporary equipment which are needed for field tests.

The Bidder shall disconnect all temporary equipment after finishing the field test.

The Bidder shall fill the systems and equipment to be hydro tested by him with water. After testing is completed, the Bidder shall drain the water and dispose of it, as directed by Owner. The water used for hydrostatic testing will be demineralized water, condensate, service water or construction water, as set forth in the Specification, and/or as directed by Owner.

4.2.8.11 Gagging During the Leak Test

In conjunction with leak testing of the piping, the Bidder shall do all gagging of associated safety and relief valves required for the leak testing and shall remove gagging at the completion of the testing.

4.2.8.12 Run Test

After the piping system and all equipment installed by the Bidder have been put in service, the Bidder shall run a check test to ascertain that the system and all the appurtenances are performing satisfactorily.

4.2.8.13 Adjustment and Tightness

The Bidder shall remedy all leaks in the piping system and appurtenances erected by the Bidder that may develop during testing and initial operation and shall retain the necessary field crew on site after the main erection work is completed to perform, the following work as required to prepare the piping system for continuous operation:

- Recheck all bolted assemblies where necessary.
- Make those adjustments on piping required by the manufacturer of apparatus to which the piping connects in order to relieve stresses and other conditions which arise after heating-up of the pipelines.
- Re-adjust tension of bolted assemblies in accordance with temperature rise.
- Adjust pipe supporting elements.
- Tighten packing glands of all valves and pumps. Additional valve gland packing rings shall be added to all valve glands, if necessary, so that a proper seal is assured and all valve glands are in correct working condition after placing in service.

4.2.9. INSULATION

4.2.9.1 General

Insulation shall not be applied to piping until all piping tests have been completed, and subject to the approval of the Owner. Insulation shall not be applied on flanged joints until approval is given by the Owner.

Before insulation is applied on any piping or equipment, the surfaces to be insulated shall be cleaned, and wire brushed if necessary, to remove any accumulation adhering to the surfaces. Additionally, any surface where adhesive is used as a means to support the insulation, shall be free of all oil film.

It shall be the Bidder's responsibility to ensure that all insulation and jacketing is properly applied and secured in a suitable manner as required for each particular purpose.

If any tank or equipment bearing a code stamp is to be insulated, the code stamped area shall contain a removable section of insulation, suitably labeled, to make code marking easily accessible for inspection.

Welding of clips, welding studs etc. is not permitted on tanks, filters or other vessels having rubber lining, or other special linings or coatings which may be damaged by the heat of welding.

The Bidder shall route all piping so that no reduction in insulation thickness is required due to proximity of pipes to conduits, steel supports, other piping etc. The Bidder shall obtain approval from the Owner, prior to any intended reduction in insulation thickness.

4.2.9.2 Materials

All insulating materials, adhesives, finishes, tapes, lagging materials, insulation sealants, mastics, vapor barriers etc. shall be received at the site in the manufacturers' unopened cartons, and shall be kept dry and protected against the elements during storage and during all stages of application.

The Bidder shall not install any cracked or damaged insulation. Insulation which is cracked or damaged during its installation shall be removed and replaced by the Bidder.

4.2.9.3 Multiple Layer Insulation

Multiple layer insulation shall be applied and installed as specified on **Part 4 Section 3, M-003**.

Staggered joint construction shall be used for joints in both directions on all multiple layer insulation.

4.2.9.4 Insulation Fastening Wire

Fastening wires for pipe and equipment insulation shall be in accordance with the following:

- Single-layer insulation and the outer layer of insulation for all double-layer applications shall be fastened with 1 .6 mm or heavier galvanized steel wire.
- The inner layer of insulation for all double layer application shall be fastened with 1 .6 mm or heavier stainless steel wire.

In addition to the above requirements, any fastening wire coming in contact with surfaces over 260°C shall be 1.6 mm or heavier stainless steel.

4.2.9.5 Joints

All insulation shall be laid with tightly butted joints and all voids, chipped corners, or other openings in the insulation sections or blocks shall be filled, not merely pointed, with an approved insulating cement, finished smooth and neat.

All forms of insulation shall present a smooth even surface so that no joints or wire grooves will appear on the finished surface.

Longitudinal joints in metal jacketing shall be lapped not less than 50 mm. Circumferential joints in metal jacketed piping and equipment shall be lapped not less than 75 mm.

4.2.9.6 Insulation for Hot Piping

Insofar as possible, all piping insulation shall be applied in sectional form. When the segmental form is necessarily used, segments shall fit the curved surfaces to which they are applied.

Not less than 3 insulation tie wires shall be applied on each 1 m long section of sectional pipe insulation. All fastening wire shall be at right angles to the axis of the pipe, not diagonally wound on. End tie wires shall be applied 75 mm from the end of each insulation section. Ends of tie wires applied to outer layers shall be tightly twisted together, bent over and pressed into the insulation so as to leave no projection visible on the finished surface.

Each layer of insulation shall be fastened independently of other layers.

Insulation on vertical runs of pipe shall be supported to prevent displacement due to slipping or contraction. Certain piping systems, will be provided with insulation support lugs and insulation support rings furnished in place. The Bidder shall provide friction clamps or shall attach clip angles, studs etc. to such other vertical runs of pipe as may be required to properly support the insulation. Friction clamps, clip angles, studs etc. and method of attachment shall be as approved by the Owner.

4.2.9.7 Insulation for Hot Flanges

On all insulated pipelines where flanged joints occur, the Bidder shall install a short section of insulation next to the flange on either side of the joint, so that this section may be removed and the bolts taken out of the joint without disturbing the insulation on the remainder of the pipeline.

Flange insulation shall, where possible, be made up of sectional pipe insulation in two halves. For flanges so large that sectional insulation cannot be used, the covers shall be built in the field with a suitable wire mesh reinforcement to which the insulation material shall be firmly fastened. The outside 15 mm of the flange covers shall be composed of cement. These flange covers shall also be made up in two halves where practical. The thickness of the insulation on the edge and sides of the flange shall be not less than the thickness of the insulation on the pipeline in which that occur.

4.2.9.8 Insulation for Hot Valves and Fittings

Insulation sections or segments used for covering valves, fittings etc. shall be properly mitred as required and shall fit together snugly. Insulation shall be properly secured by tie wire as hereinbefore specified.

Insulation for valves and fittings where the O.D. of the insulation is 150 mm or greater shall be reinforced with 25 mm mesh, 1.25 mm galvanized wire netting and finished with a 15 mm thick, when dry, coating of hard finish cement. Netting shall not be visible through cement finish. Insulation for valves and fittings made in accordance with ASTM Specification C450 will not require mesh reinforced cement over the insulation.

4.2.9.9 Removable Covers for Valves and Fittings

Removable, reusable covers for valves, flanges etc. where necessary for maintenance purposes, shall be of the following types:

- Insulated Cans
- Metal Mesh Encased Blankets
- Fabric Encased Blankets

4.2.9.10 Insulation for Hot Equipment

Insulation for tanks, heaters, and other equipment with removable heads shall be applied so that the insulation can be removed from the heads, and the heads can be removed without disturbing the shell insulation.

The shell flanges and head flanges on such equipment shall be covered with insulation of the same thickness as specified for the shell. The insulation shall be applied on head flanges so that it can be removed and replaced in section without undue difficulty and breakage.

Manholes or handholes in insulated tanks and equipment shall be provided with a removable, reusable insulation section of sufficient size to expose the bolts or fastening devices of the manholes and handholes when the section is removed. Removal and replacement shall be accomplished without damaging the removable insulation section or the adjacent insulation.

Method of insulating head flanges, manholes and handholes shall be as approved by the Owner.

The Bidder shall furnish and install welded-type studs on the heads, lower halves and vertical sides of cylindrical heaters, tanks etc. for attaching insulation holding wires, to hold insulation securely in place independent of any circumferential wires that may be used.

Studs shall be spaced on a not greater than 450 mm centre both ways, and on a closer centre where required to ensure proper attachment of insulation.

All sectional, segmental or block forms of insulation shall be securely held in place by means of wire. Each layer shall be fastened independently of other layers. Wire shall be secured in such a manner that no projections will be visible through the insulation finish.

All block or segmental insulation on equipment, not enclosed in metal casings or lagging shall be covered with 25 mm hexagonal mesh, 0.90 mm galvanized wire netting for indoor applications and 1.25 mm galvanized wire netting for outdoor applications.

Separate tie wires shall be used for attachment of netting. These wires shall be anchored to the insulation fastening studs and not to the insulation tie wires.

After the wire netting is securely held in place, the entire surface shall be finished with hard setting cement, and to a smooth even surface. Cement finish shall be 15 mm thick when dry.

In lieu of the tie wires and welding studs indicated above, calcium silicate blocks may be secured by means of 0.5 mm thick, 15 mm wide, stainless steel metal bands placed on not greater than 300 mm centers for equipment up to and including 1220 mm O.D.

4.2.9.11 Personnel Protection

Personnel protection insulation shall be provided as specified in **Part 4 Section 3, M-003**.

Piping exterior to the building requiring personnel protection shall be insulated.

The above drains and vents shall be insulated with the same thickness of calcium silicate insulation, applied, attached and finished as specified for other associated piping of the same operating temperature.

4.2.9.12 Insulation for Temporary Piping

Portions of the temporary piping shall require insulation in accessible areas, as outlined above for personnel protection and as specified in **Part 4 Section 3, M-003**.

Metalcladding shall be required over the insulation of temporary piping located outdoors (if any), to keep the insulation watertight.

4.2.9.13 Cladding

Piping under the turbine and for a distance up to 3.00 m beyond the perimeter of the foundations, on all levels below the main floor shall not have a moisture barrier applied under the aluminum cladding. Aluminumcladding coated with epoxy on the inside is permitted in this area.

Cladding for elbows shall be preformed metal elbows or factory- made gored sections. In sizes where preformed metal cladding or factory-made gored sections are not obtainable, elbows shall be lagged using carefully fitted mitered lagging sections. Not less than 3 screws shall be used on the inside or outside radius of each preformed metal elbow, but in no case shall be the spacing of screws on the inside or outside radius exceed 150 mm.

Metalcladding shall be tight, and where used outdoors, shall be weathertight under the most severe weather conditions.

Longitudinal joints in metal cladding shall be lapped not less than 50 mm. Circumferential joints on metal-clad vertical pipes shall be lapped not less than 50 mm, circumferential joints in metal-clad horizontal pipes shall be lapped not less than 75 mm.

Allhorizontal joints in metal cladding shall be turned downward. Laps in metalcladding for outdoor piping shall be located 45° below the horizontal centerline of the pipe.

Cladding where required over vaporproofed surfaces shall be held in place by 12.5 mm wide stainless steel bands placed on not greater than 300 mm centers.

All other cladding shall be secured with suitable self-tapping sheet metal screws. Screws shall be placed on not greater than 150 mm centers.

Metal cladding for equipment and flanges, manholes etc. shall be designed to be removed and replaced without damage to insulation or cladding.

Metal cladding for outdoor surfaces shall be caulked at all penetrations to make such surfaces weathertight.

4.2.9.14 Canvas Jacketing

All canvas shall be field-applied. Factory-applied canvas is not acceptable.

Before the canvas jacket is applied, care shall be taken to ensure that all joints and irregularities on the insulation are properly pointed up and filled with cement to present a smooth surface for the canvas.

The canvas adhesive shall be mould inhibiting type and shall be tinted (light blue) so that it can be readily determined that the canvas is completely coated. Color density shall be as approved.

In applying canvas covering, it shall be cut so as not to come in contact with any hot metal surfaces such as pipe supports, flanges etc.

Where metal surfaces at edges of insulation are in excess of 90°C a 150mm wide strip of fiber glass cloth shall be pasted on over the insulation instead of the canvas. This shall apply where the canvas would make contact with metal having operating temperatures above 90°C such as valve bonnets, flanges, piping and equipment.

Fibre glass cloth shall be pasted on with silicate of soda.

Canvas shall lap the fibre glass cloth not less than 50mm wherever fibre glass cloth is used.

4.2.9.15 Weatherproof Coating and Jacket

Weatherproof coating shall be applied to a dry thickness of not less than 4.8 mm. Application shall be in accordance with manufacturers' instructions. Color of material shall be gray.

Before applying the weatherproof coating the entire insulated surface shall be covered with 25 mm hexagonal, 1.25 mm galvanized wire mesh, galvanized after forming, for reinforcement of the coating. Reinforcement shall be securely attached to the insulation fastening devices by means of 2 mm galvanized steel wires or speed washers if pins are used. Speed washers shall be not less than 25.6 mm. Pins shall be bent over and not cut. Reinforcement shall be lapped not less than 75 mm at all edges. Not less than 2.00 mm galvanized steel wire shall be used to tie together. Lacing will not be permitted.

Wherever weatherproof coating comes in contact with temperatures in excess of 8 0°C, and at all penetrations, insulation shall be flashed with suitable mastic having a temperature use limit of not less than 150°C. Mastic shall be reinforced with 250 x 250 mm nonasphaltic glass cloth and shall extend not less than 150 mm back from the hot surface. Colour of mastic shall be black.

All corners shall be straight and true.

All longitudinal and circumferential joints in the weatherproof jacket shall be lapped at least 75 mm and all horizontal laps shall be located on the side of the pipe 45 degrees below the horizontal centerline with the laps turned downward in order to shed water. All joints shall be properly sealed with an approved

weatherproof cement. The jacket shall be secured in place by means of rings of 1.6 mm copperclad steel wire applied on not greater than 150 mm centres.

All insulated pipe surfaces where the weatherproof jacket cannot be conveniently applied, such as on flanges, fittings, valves etc. shall be covered with 25 mm aperture mesh 1.25 mm hexagonal netting, galvanized after forming. After the mesh is securely held in place weatherproof coating as approved shall be applied over these surfaces. Weatherproof coating shall be 6.4 mm thick when dry.

4.2.10. FLUSHING AND STEAM BLOW

In interpreting the following requirements, no plant items incorporating copper or aluminium or their alloys, or zinc coatings shall undergo any chemical cleaning other than flushing or manual cleaning as agreed with the Owner.

The Bidder shall take all necessary precautions to ensure that the internal surfaces of all plant are kept clean and free from injurious matter during erection.

When all plant has been erected and lagged or at such other time as may be agreed with the Owner for sub-assemblies, the installation shall undergo a procedure for Site cleaning in accordance with the requirements and associated Schedules as specified herein.

4.2.10.1 Piping and Vessels

All piping and vessels shall be flushed with air or oil, or steam or deionised, or distilled, or potable, or sea water, as deemed appropriate by the Owner to remove loose debris.

4.2.10.2 Feedwater Systems

Piping and vessels on the water side of plant between the steam turbine condenser and boiler but excluding pumps shall first be flushed as detailed above by the Bidder after which the following procedure shall be carried out .

The boxed up system shall then be sealed with nitrogen in the event that starting time, prior to commissioning will exceed 30 days.

The Bidder shall be required to include in his proposal but not be limited to for the following, as applicable:

- Supply and erection of suitable bypass pipework around the boiler feed pumps.
- Isolation or removal and replacement of instrumentation as necessary.
- Provision of personnel and all necessary equipment during cleaning procedure.

4.2.10.3 Oil Systems

Before any machine is run on Site the oil system shall have been thoroughly cleaned and flushed. The complete system shall be flushed, excluding such items as bearings and control valves etc. Temporary strainers shall be incorporated in the flushing arrangements to collect foreign matter and debris, and to indicate when a satisfactory state of cleanliness has been achieved.

All unalloyed ferrous lubricating oil pipes shall be pickled.

The site flushing procedure shall be submitted to the Owner for approval. The oil shall be circulated in the flushing procedure at a temperature satisfactorily in excess of any emergency have appeared free from debris the circulation of flushing oil shall continue for a further three hours, and the strainers again checked to confirm satisfactorily cleanliness of the system. All flushing oil, temporary pipework, strainers, and any other materials required for the Site flushing procedure shall be supplied under this Contract. Both hot and cold flushing shall be carried out.

4.2.10.4 HRSG (except for Superheaters, Steam Piping and Pumps)

The HRSG shall be flushed with potable water or equal quality to remove loose debris.

The HRSG shall then be filled to an agreed cleaning level with potable water or equal quality, containing 1 gram/litre trisodium phosphate. The pressure shall be raised to 43 bar g or normal working pressure, whichever is the lower. The pressure shall be reached in 6 to 10 hours, and maintained for 24 hours. The HRSG shall then be allowed to cool and blowdown to empty when pressure has decayed to 12 bar g.

During the boil out, all blowdown and drain connections shall be operated for 30 seconds each hour.

The drums and headers shall be opened up and any heavy loose debris removed.

The HRSG shall then be flushed with water of potable or equal quality to remove loose debris.

The HRSG shall then be filled to an agreed cleaning level with deionized or distilled water and the temperature raised as required. If the Bidder considers that only 100% deaerated deionized water should be admitted to the HRSG on any occasion following boiling-out whenever steam pressure and quantity cannot be made available from the auxiliary steam header, he shall provide an external start-up steam supply to operate the deaerator for heating, recirculation and deaeration via the feed storage vessel.

In this respect, although the HRSG may utilise the gas turbine waste heat to raise or supply the necessary steam, it may not be possible to pressurise the main steam lines due to work in progress on the steam turbine.

The HRSG shall be placed into service within the minimum possible time after inspection.

In the event that the delay in commissioning exceeds one week but is less than four weeks, the passivation procedure shall be repeated. In the event that the delay in commissioning exceeds four weeks the HRSG shall be either dry stored with suitable desiccants in trays, or nitrogen sealed or wet stored by filling to the air cocks, including drums and superheaters where practicable, with a solution of deionised water of 50 mg/l hydrazine, 0.2 kg/l copper (added as copper sulphate) and ammonia to raise the pH value to not less than 10.5.

The Bidder shall state clearly in the proposal the suitability of the above procedure for the HRSG offered.

4.2.10.5 Superheaters and Steam Piping - Steam Blow

At the completion of the passivation stage the HRSG pressure shall be adjusted as necessary and the steam cleaning procedure commenced.

The steam cleaning procedure for each section of the works shall ensure that the superheaters and the high pressure steam pipework between the HRSG and the turbine (including also the cross-over pipework between the two sections and the each pressure steam bypass steam) are suitably blown through with steam from the HRSG to ensure adequate cleanliness of all internal surfaces. The steam conditioned and velocities achieved during individual steam blows shall produce a scouring effect on the tube and pipe walls at least 60% in excess of the value obtaining during normal service (a drag factor greater than 1.6.).

$$\text{Drag Factor} = \frac{(\text{mass flow} \times \text{velocity}) (\text{steam blow})}{(\text{mass flow} \times \text{velocity}) (\text{MCR})}$$

Temporary pipework and supports shall be provided by the Bidder to take the steam away from the turbine and condenser inlets to a safe place of discharge and, if necessary, silencing of the steam outlet pipe shall be provided. The temporary pipework shall be fitted with provision, for fitting target plates for sampling the cleanliness of the steam, and the results of such sampling shall establish to the satisfaction of both the Owner and the Bidder, the number of steam blows required.

During steam blowing all make-up to the steam generator shall be treated with 50 mg/l ammonia and 50 mg/l hydrazine.

To minimise risks of stress corrosion to superheaters arising from carry-over, no caustic soda shall be added to the HRSG water for the steam blow.

The HRSG water shall not contain more than 2.0 mg/l sodium chloride at any time during the blow.

At the end of the steam blow, the plant shall be allowed to cool and the pressure to decay. The plant shall be blown empty when pressure reaches 6 bar g.

The HRSG and any other appropriate sections of the plant will then be opened up for inspection.

The Bidder shall co-operate fully with the Owner during all stages of the steam blow programme.

The Bidder shall provide all blanking arrangements necessary for the steam chest internals to adequately prevent admission of steam to the turbine cylinders during the steam blow programme.

It is not intended to carry out a steam blow of the Low Pressure bypass pipe unless the Bidder wishes to do so, subject to the agreement of the Owner. However the Bidder is to make provision to satisfy the Owner that this pipeline is completely clean after erection. This may be done by the use of mirrors, intrascope or closed circuit TV procedures or other method to be agreed with the Owner.

4.2.10.6 Disposal of Waste Effluent

The Bidder will be responsible for satisfactorily disposing of all waste effluent liquors and materials and treating them to render them suitable for discharge to waste in accordance with regulations prevailing at Site regarding disposal of waste effluent.

4.2.10.7 Inspection after Cleaning

At the end of the plant cleaning operations at Site the plant shall be made available for inspection.

In the event that the surfaces are not cleaned to the Owner's satisfaction such parts of the cleaning procedure or agreed alternatives as are deemed necessary to overcome the deficiencies shall be carried out at the Bidder's sole expense.

Following complete erection of the turbine, main steam pipework, turbine bypass pipework and other pipework that is not subject to the steam blow procedure, then such pipework that may have debris lodged therein that may be injurious to the turbine, shall be internally inspected by means of mirrors, intrascope, or CCTV procedures or other methods to be agreed by the Owner.

4.2.10.8 Submittal for Cleaning Procedure

At a time not less than six months prior to commencement of any site cleaning, the Bidder shall submit a detailed site cleaning specification and programmed including procedure details, supporting calculations giving cleaning times, velocities, chemical solutions if applicable, silencing, drag factors, pipework forces and moments imposed during site cleaning operations.

4.2.11. FIELD TESTS FOR MECHANICAL EQUIPMENT

4.2.11.1 HRSG

4.2.11.1.1 Hydrostatic Test

After erection a hydrostatic test shall be carried out on HRSG with all mountings, fittings and associated piping in compliance with the relevant standard approved by the Owner with all fittings and mountings in position. No control valve shall be used as a block valve for the purpose of this or any other test.

4.2.11.1.2 HRSG Setting Smoke Test

Smoke test shall be performed prior to commissioning of the HRSG and prior to the fixings of the external insulation and/or cladding to demonstrate the gas tightness of the boiler casings and associated fittings and flues.

The gas duct shall be temporarily blanked off at the inlet to the stack and the smoke test carried out with an air compressor. Alternatively, and subject to the Owner approval, separate tests may be carried out on the inlet duct, and outlet duct using compressors. The source of the smoke shall be distributed within the pressurised volume and shall be producing smoke throughout the pressurising period.

Defects shown by this test shall be made good and proved by a further smoke test, at the Bidder's sole expense, before the external casing of the HRSG is lagged and/or clad.

4.2.11.1.3 Storage of HRSG

Should it be necessary to store a HRSG, the Bidder shall be responsible for ensuring the HRSG is maintained in a passive state until it can be put into commercial operation. The Bidder shall provide chemicals, etc., necessary to maintain the HRSG in a non-corrosive state in accordance with the clause on HRSG cleaning. Demineralised water and nitrogen for storage of HRSG shall be provided by the Bidder.

4.2.11.2 Field Fabricated Tanks

All site fabricated tanks shall be subject to a hydrostatic pressure test in accordance with the relevant standard for a minimum of 30 minutes.

Tanks operating at atmospheric pressure shall be tested by fillings to maximum level and being subjected to an additional superimposed hydrostatic pressure of 1.0 bar(g). The external surfaces shall be thoroughly chalked and the pressure maintained for a minimum of 24 hours.

4.2.11.3 Condensing Plant

The condensers are to be given a fluorescent water test after tubing and before fitting the water boxes.

After completion of tubing and fitting of the waterboxes, the C.W. sides of the condenser between the inlet and outlet isolating valves shall be subjected to a hydrostatic pressure test to pressure of 1.5 times the system design pressure.

The condenser shall be filled with water to the level of the lowest L.P. turbine blade. The casing shall be thoroughly soaped and the water level maintained for two hours to check for leakage.

The internal rubber linings of the waterboxes and doors shall be tested for satisfactory adhesion by a spark test method over the complete surface of application.

4.2.11.4 Air-Conditioning and Ventilations

4.2.11.4.1 General

A comprehensive report shall be prepared for the ventilation and air conditioning system, which shall record all the parameters which affect the performance of the system at the time of commissioning. A similar report shall be prepared for specified tests. The reports shall be submitted to the owner for approval and such approval must be gained before the services are handed over to the Owner.

4.2.11.4.2 Air Distribution and Extract System

All distribution and extract systems shall be commissioned in accordance with the CIBS (IHVE) Commissioning Code Series A, Air Distribution System, High and Low Velocity.

4.2.11.4.3 Automatic Control System

All automatic control system shall be commissioned in accordance with the CIBS (IHVE) Code Series - The Commissioning of Automatic Control System and the control equipment Supplier's recommendations.

4.2.11.4.4 Refrigerating Systems

Refrigerating systems shall be commissioned in accordance with the CIBS (IHVE) Commissioning Code Series R - Refrigerating System and to the equipment Supplier's recommendation.

4.2.11.4.5 System Performance Tests

Air conditioning system performance tests shall be carried out which shall be witnessed by the Owner. The tests shall comprise:

Demonstration that the systems are capable of achieving the specified conditions inside buildings.

Measurement of pressure drop through filters.

Measurement of air temperature drop through the cooling coils at the specified air volume flow rate.

Measurement of the outside air condition every two hours and at the beginning and end of each testing period.

Measurement of the following temperatures and pressures and a record of the time when the measurements were made.

- Compressor discharge pressure and temperature
- Air entering and leaving plant (Wet and Dry Bulb Temperatures)
- Each room (Wet and Dry Bulb Temperatures)
- Evaporator refrigerant pressure
- Condenser refrigerant pressure

Measurement of the running current of all associated electric motors.

Calibration of all installed instruments.

Demonstration that all the automatic controls are operational.

The above tests shall be carried out after the air systems have been balanced.

The Bidder shall give the Owner fourteen days notice in writing of his intention to carry out any tests.

4.2.11.4.6 Noise Level Tests

Noise levels shall be measured for each octave band with the ventilation and air conditioning systems in operation under normal running conditions.

The tests will be carried out in each room served by the air distribution systems. The test comprise of:

- Demonstration that the specified noise rating has been achieved
- Measurement of the external background noise level for each zone at the same time that the internal noise levels are measured
- Record of the state of the internal finishes in rooms

- Record of the microphone position for each measurement

4.2.11.5 Fire Protection Services

4.2.11.5.1 General

The Bidder shall arrange for pressure tests to be witnessed by the Owner. The Bidder shall submit test certificates for tests carried out as required by the Fire Department and arrange for inspections by the Fire Department for issue of Certificate of Fitness.

4.2.11.5.2 Pipework

Pipework shall be hydrostatically tested to 1.5 times the working pressure or 10 bar g whichever is higher. The test pressure shall be maintained for 24 hours without significant drop and Owner shall be notified 48 hours in advance of any testing to be carried out. During the test, screwed and welded joints shall be inspected for leakage.

4.2.11.5.3 Fire alarm and detection system

The quality of materials and finishes and the performance of all components, apparatus and equipment shall be adequate to meet the test requirements specified below and elsewhere, and to ensure satisfactory operation of the system for which the installation is designed.

Durability

Each component for the automatic and manual points shall be so constructed that it will be in good working order and undistorted after 10,000 operations.

Vibrations tests

Components shall be tested to ensure freedom from false operation or failure through mechanical shock or vibration likely to be experienced in normal service.

Insulation test

The insulation between each current carrying part and all other current carrying parts connected to any one circuit, together with all metal enclosures and metal parts of any component shall withstand the application of one minute of a test voltage of 1,000 V rms of approximately sine wave form at a frequency of 50 Hz. Immediately after this test, the insulation resistance measured at 500 V shall be not less than 20 mega ohms.

Operating test

The entire systems shall be tested to ensure that it operated satisfactorily at the normal voltage and at 80% of the normal voltage.

(Last page of Subsection 4.2)